Exercise 1

Getting familiar with LT-Spice

Winter Semester 2025/26

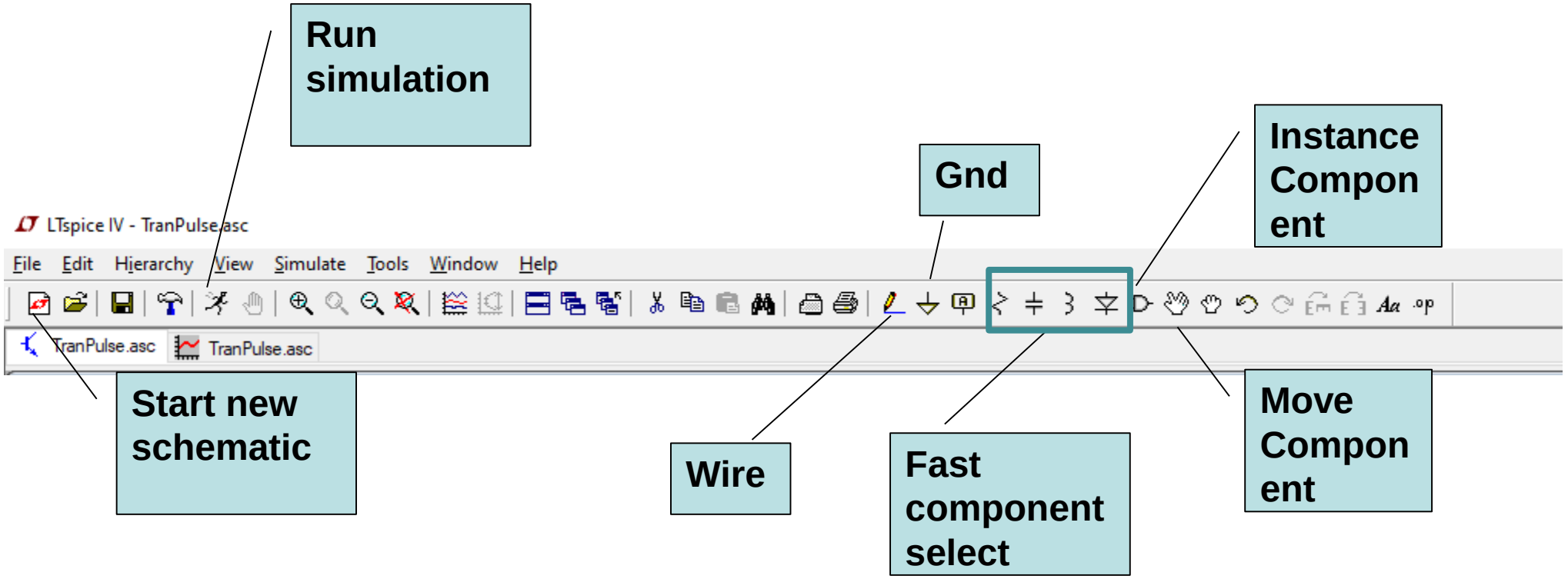
Nima Shahpari, Ph.D.

- Download LTspice IV directly from the Linear Technology homepage
  http://www.linear.com/designtools/software/

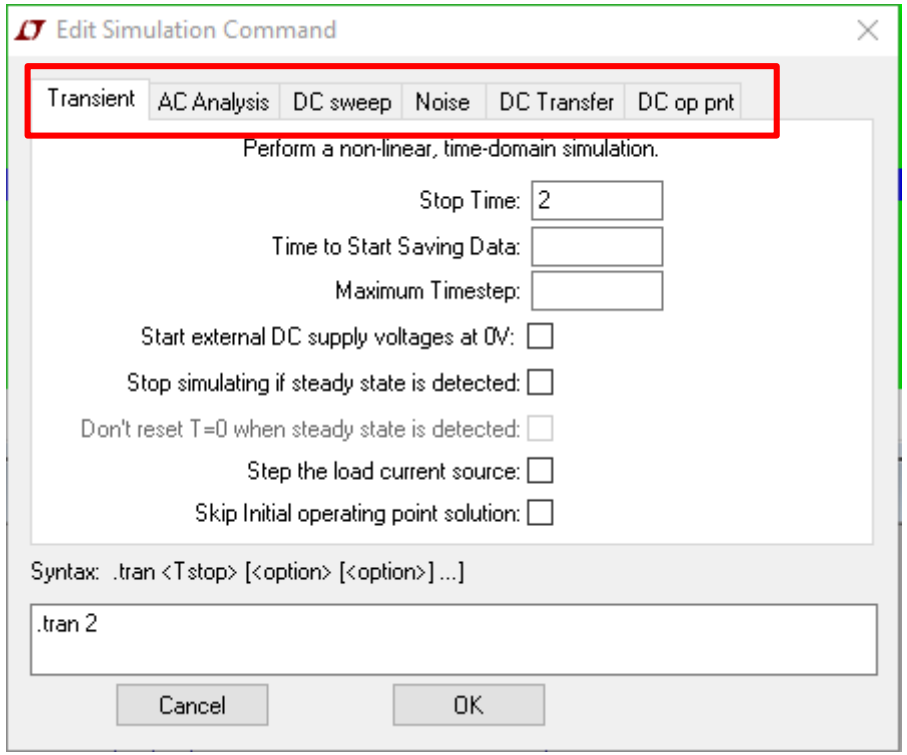
- Here, a link to a basic tutorial about LTspiceIV:

  http://denethor.wlu.ca/ltspice/

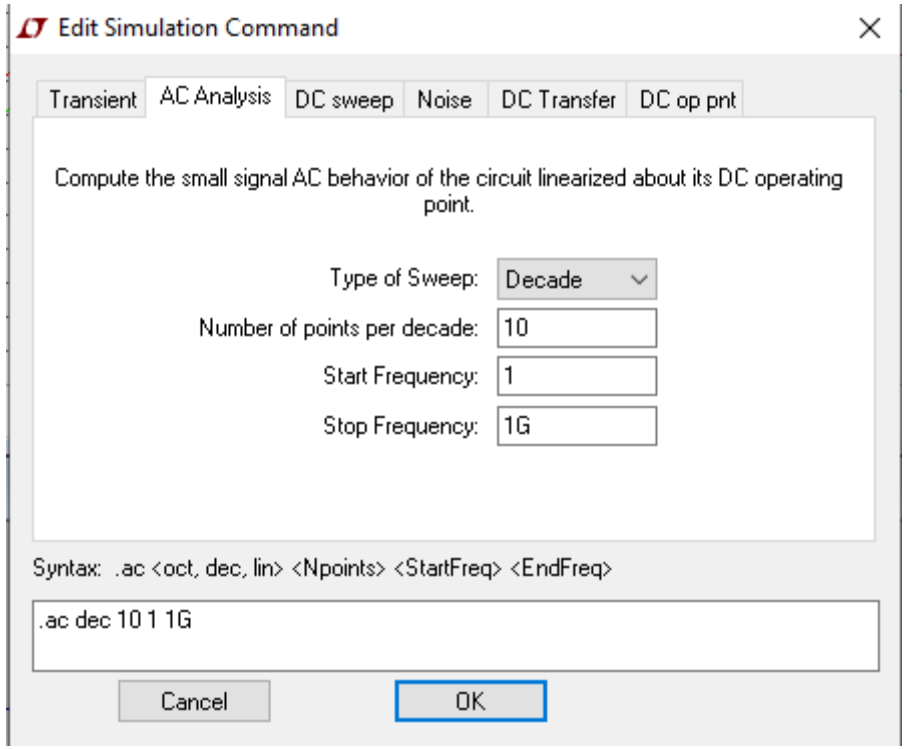
LTSPICE


Run simulation
Gnd
Instance Component
LTspice IV - TranPulse.asc
File Edit Hierarchy View Simulate Tools Window Help
TranPulse.asc
TranPulse.asc
Start new schematic
Wire
Fast component select
Move Component

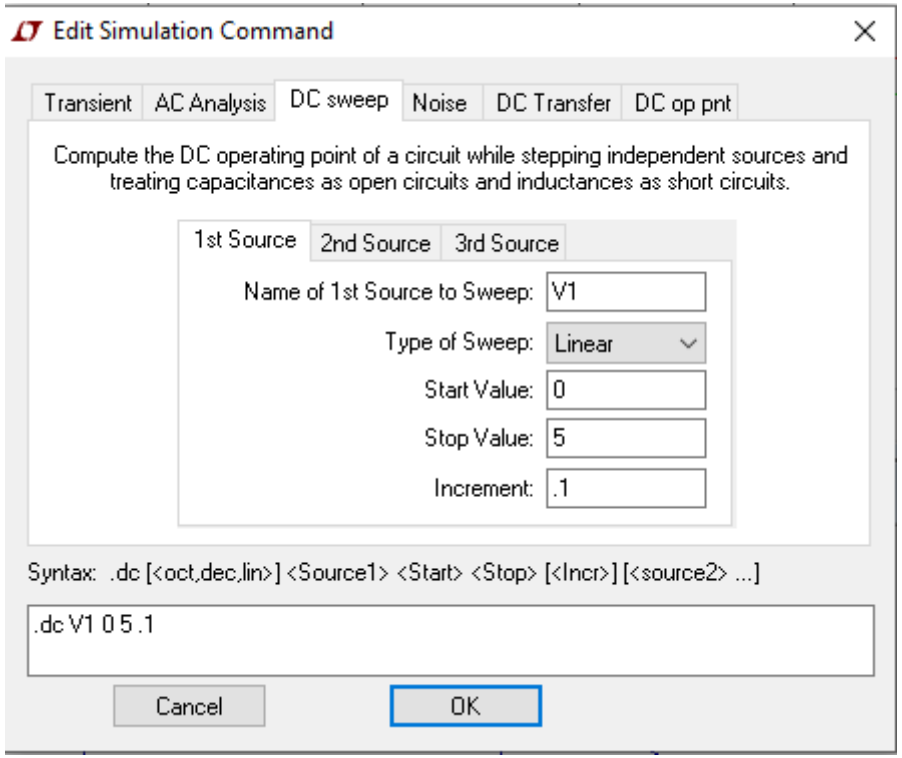
Selecting the simulation type



Selecting the simulation type



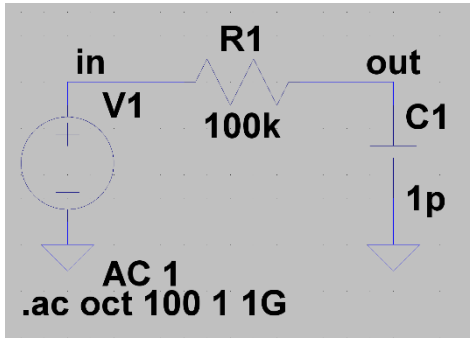
Selecting the simulation type



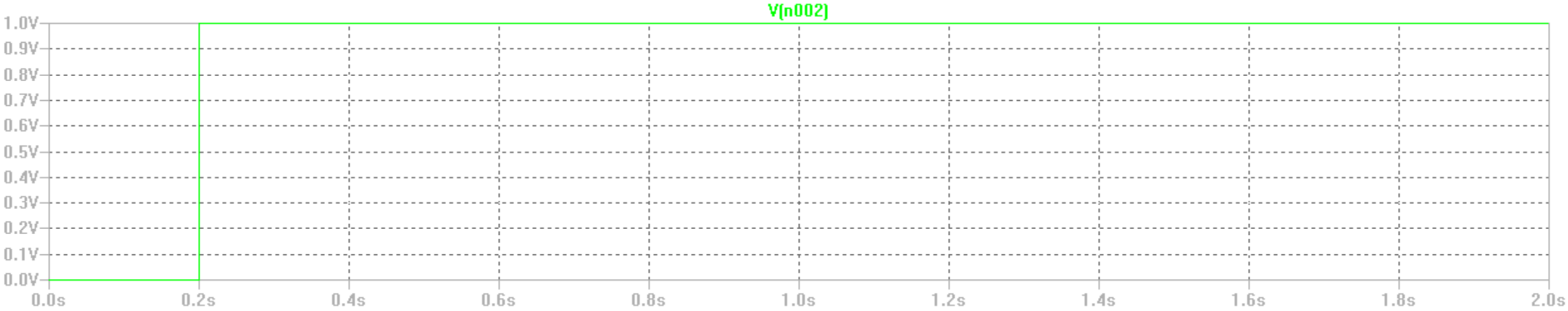
LOW PASS CIRCUIT SIMULATION

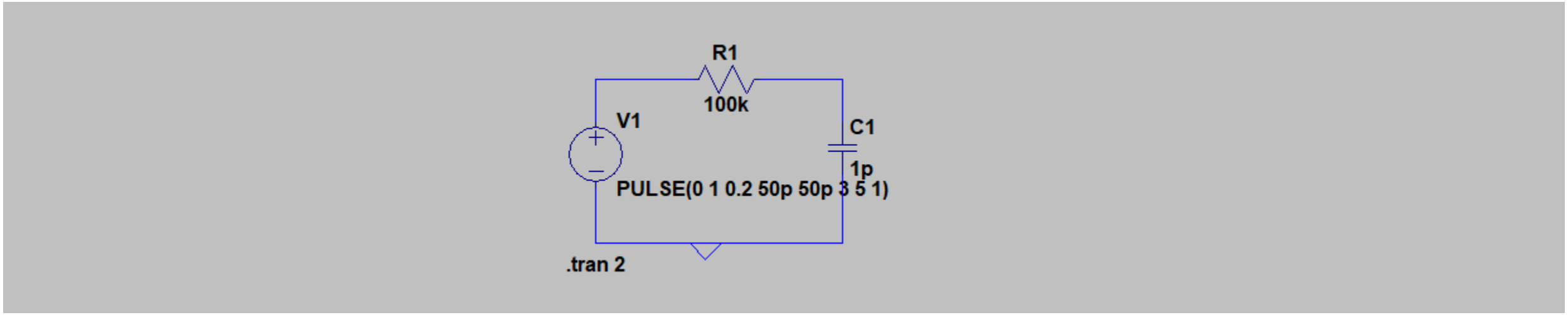
Time Domain(Transient) Simulation 1, Pulse simulation

- Vinitial[V] = 0, Von[V] = 1
- Tdelay[s] = 0.2, Trise[s] = Tfall[s] = 50p,
- Ton[s] = 3, Tperiod[s] = 5, Ncycles = 1;
- Stop time = 2 s
- Plot V(out)

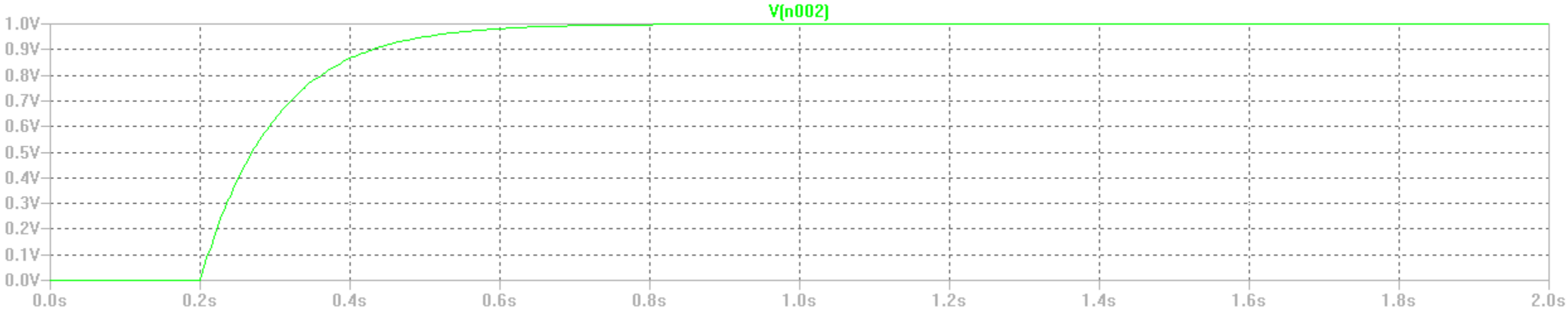


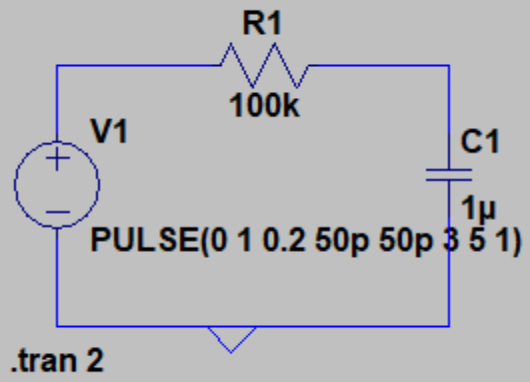
- C=1p


V(n002)
1.0V
0.9V
0.8V
0.7V
0.6V
0.5V
0.4V
0.3V
0.2V
0.1V
0.0V
0.0s
0.2s
0.4s
0.6s
0.8s
1.0s
1.2s
1.4s
1.6s
1.8s
2.0s


R1
100k
V1
C1
1p
PULSE(0 1 0.2 50p 50p 3 5 1)
.tran 2

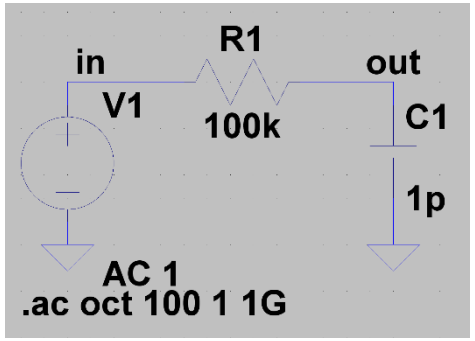
- C=1u


V(n002)
1.0V
0.9V
0.8V
0.7V
0.6V
0.5V
0.4V
0.3V
0.2V
0.1V
0.0V
0.0s
0.2s
0.4s
0.6s
0.8s
1.0s
1.2s
1.4s
1.6s
1.8s
2.0s

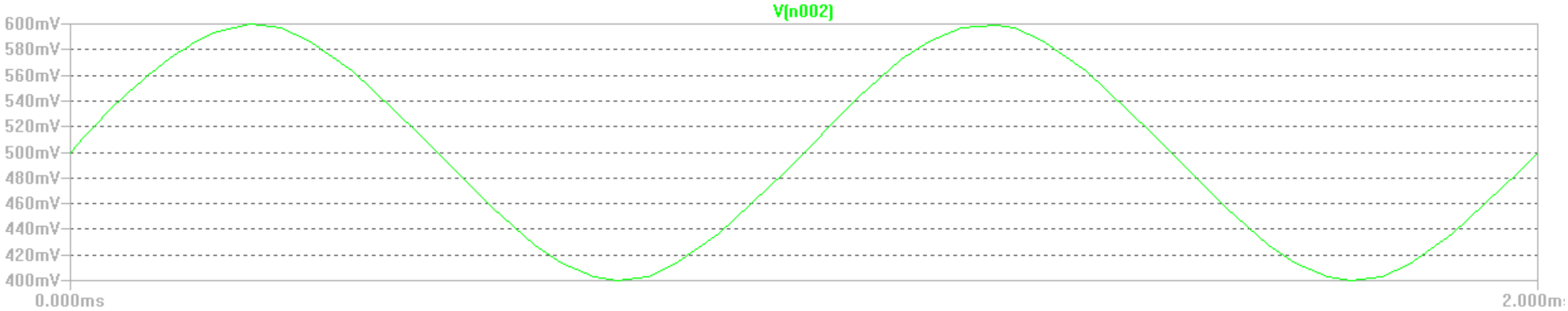

R1
100k
V1
C1
1µ
PULSE(0 1 0.2 50p 50p 3 5 1)
.tran 2

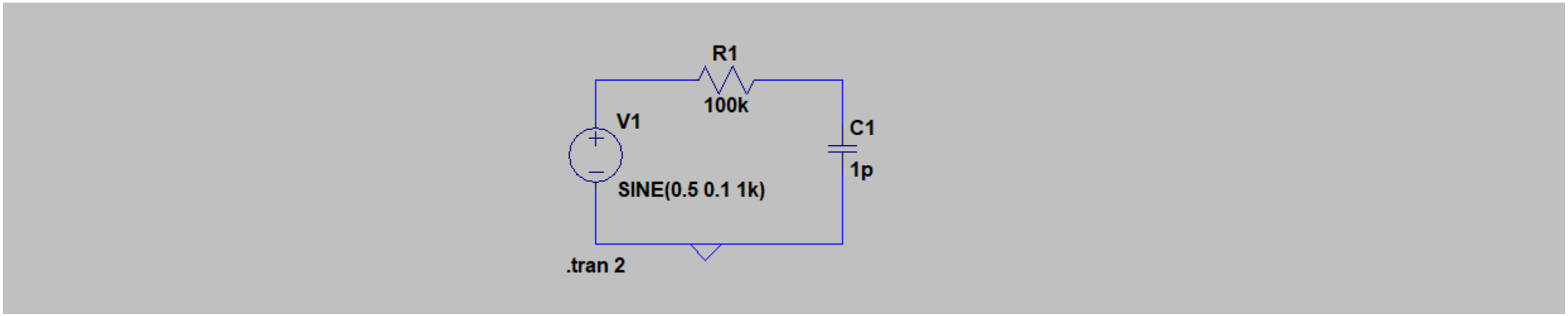
Time Domain(Transient) Simulation 2, Sine wave simulation

- Voff = 0.5, Vamp = 0.1, Freq 1k
- TSTOP = 2 ms
- Plot V(out)

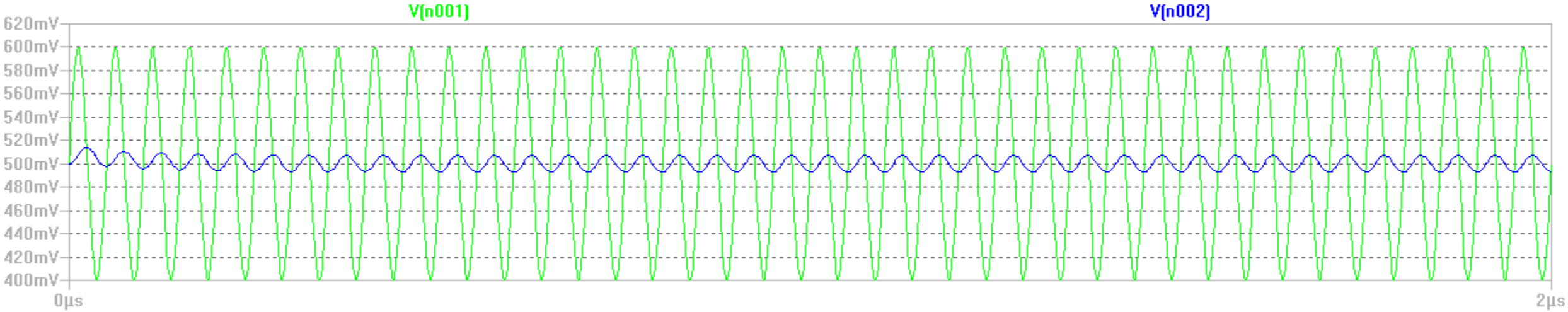


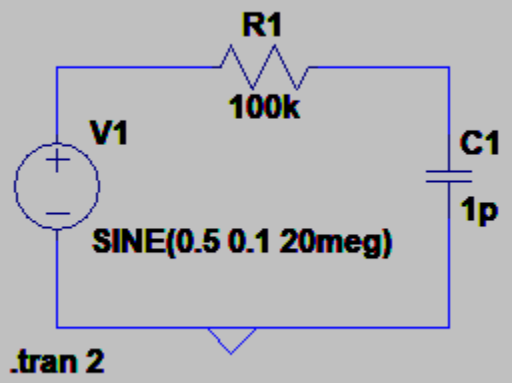
- 1KHz


V(n002)
600mV
580mV
560mV
540mV
520mV
500mV
480mV
460mV
440mV
420mV
400mV
0.000ms
2.000m


R1
100k
V1
C1
1p
SINE(0.5 0.1 1k)
.tran 2

- 20Meg


V(n001)
V(n002)
620mV
600mV
580mV
560mV
540mV
520mV
500mV
480mV
460mV
440mV
420mV
400mV
0µs
2µs

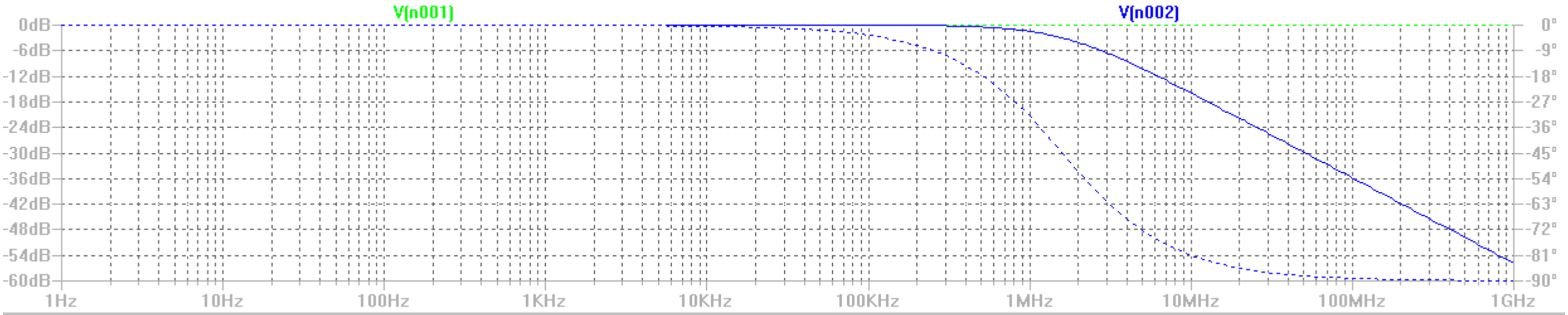

R1
100k
V1
C1
1p
SINE(0.5 0.1 20meg)
.tran 2

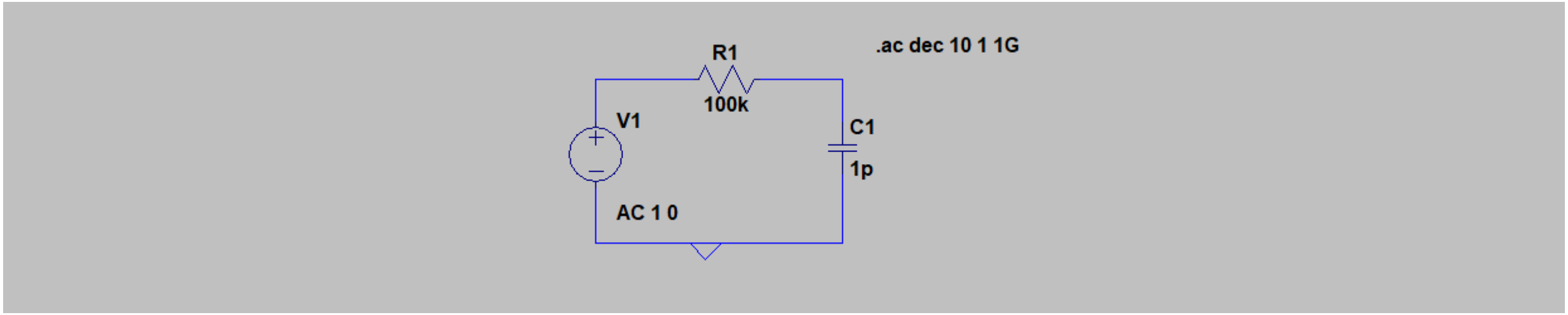
AC Sweep Simulation

- AC Sweep Simulation
  - Change V1 to AC Source (Vac = 1, Vdc = 0)
  - AC sweep:
    - Start frequency = 1 Hz
    - End frequency = 1 GHz
    - 10 points/decade
  - Plot V(in) and V(out)

AC Sweep Simulation

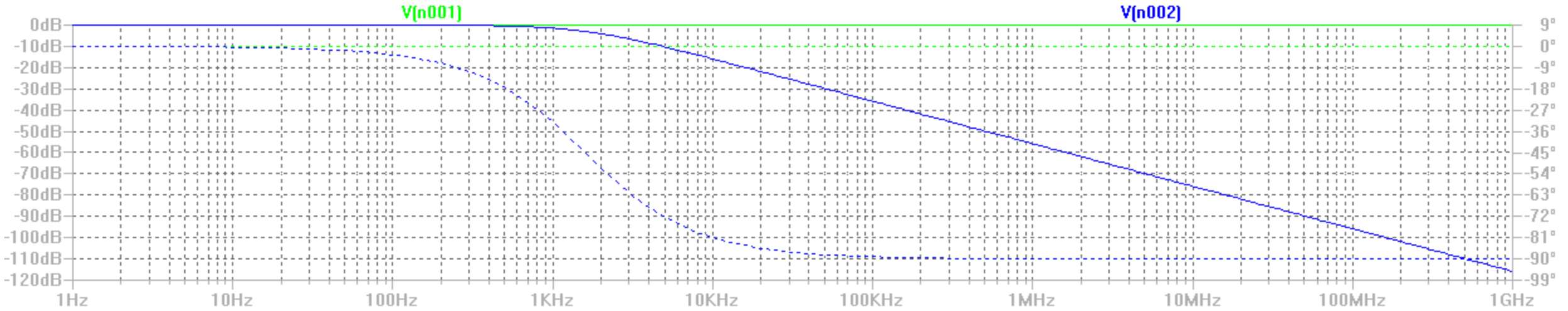
- C=1pf

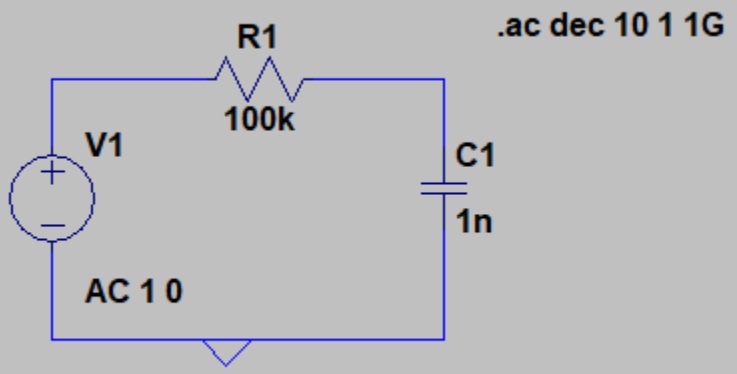




AC Sweep Simulation

- C=1nf

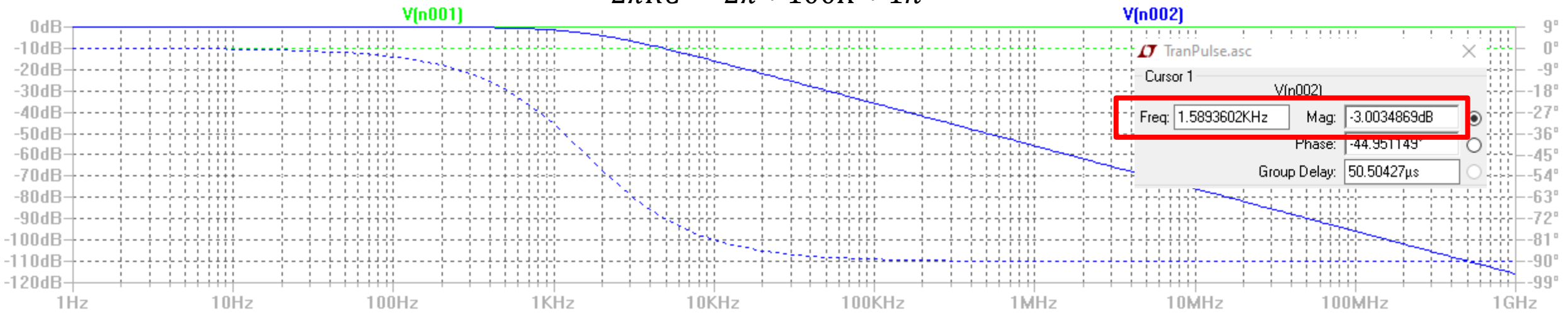




3dB Corner frequency

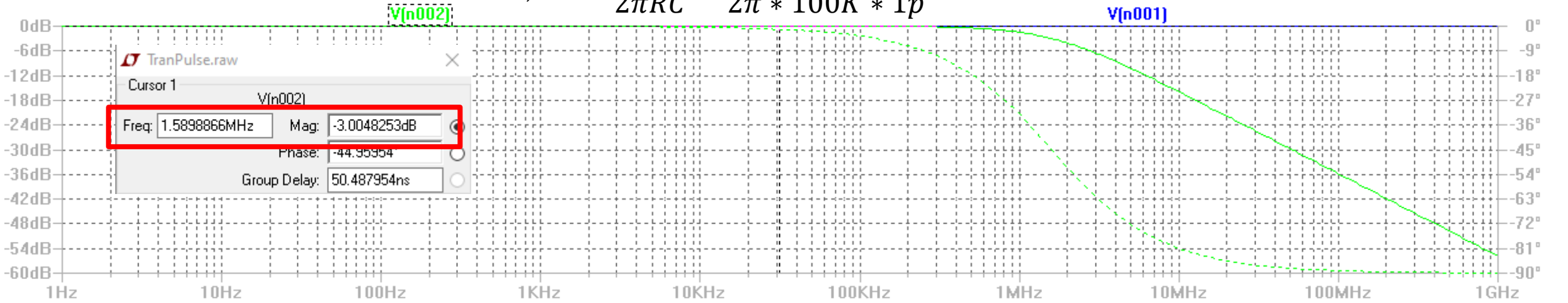
- C=1nf

$$F_{c,3dB} = \frac{1}{2\pi RC} = \frac{1}{2\pi * 100K * 1n} = 1.59KHz$$



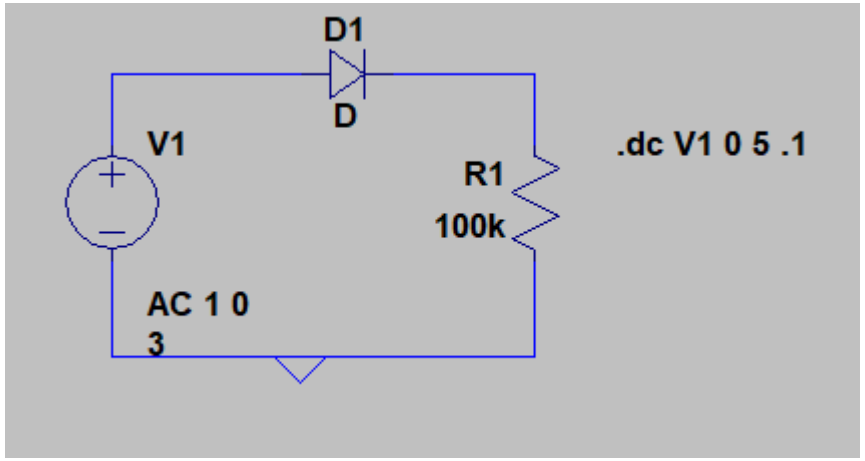
- C=1pf

$$F_{c,3dB} = \frac{1}{2\pi RC} = \frac{1}{2\pi * 100K * 1p} = 1.59MHz$$

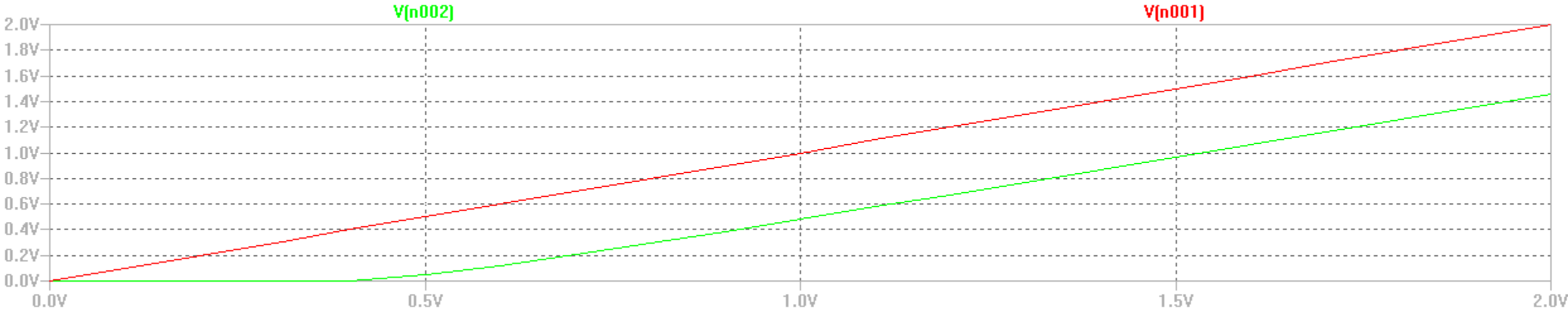


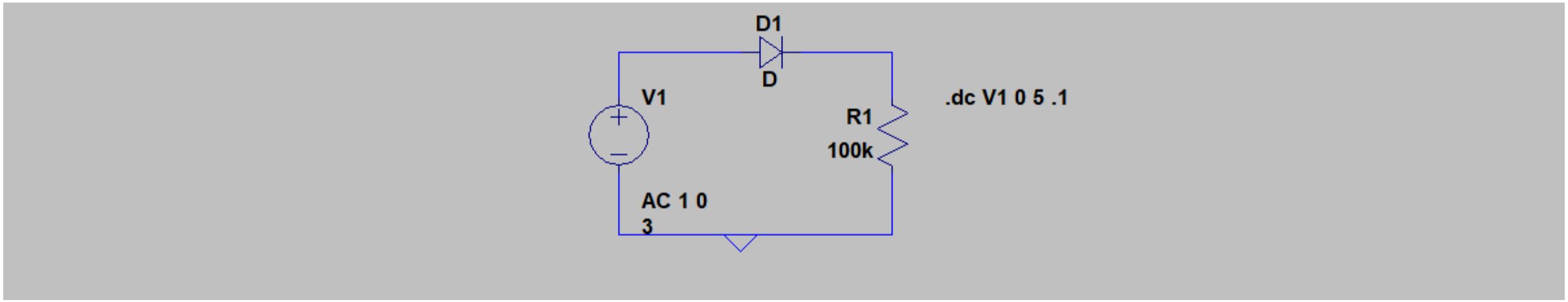
DC Sweep Simulation

- DC Sweep Simulation
  - Change V1 to DC Source (Vdc = 2)
  - DC sweep:
    - Start Value = 0
    - Stop Value= 2
    - Increment= 0.1
  - Plot V(in) and V(out)



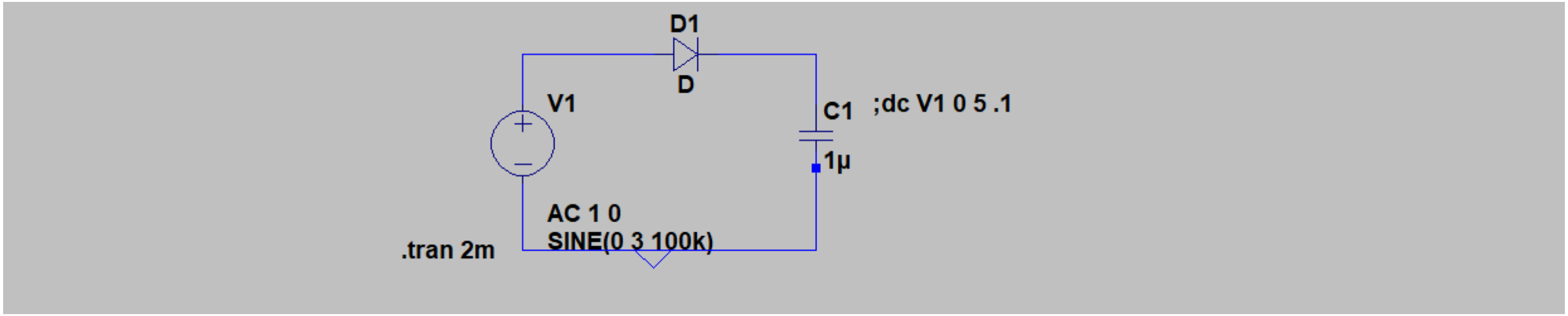
DC Sweep Simulation





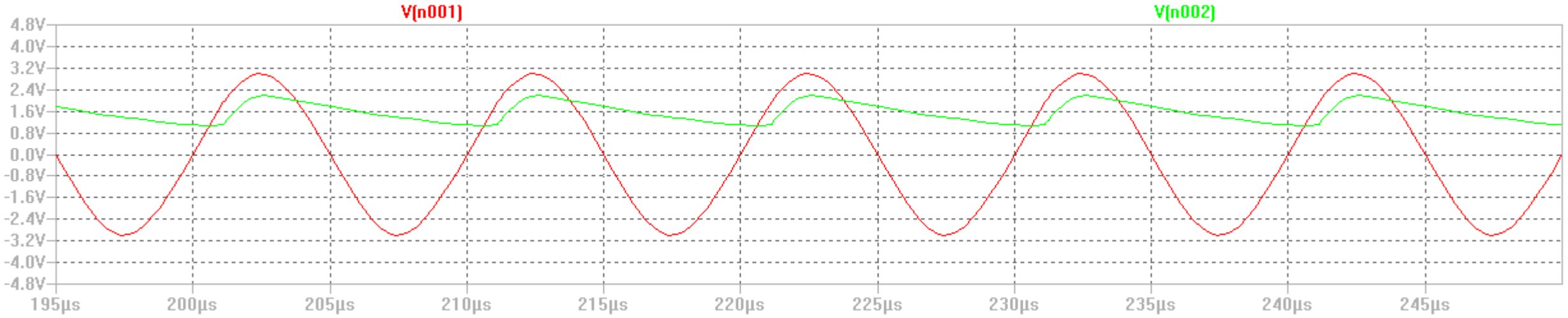
Diode-Cap rectifier

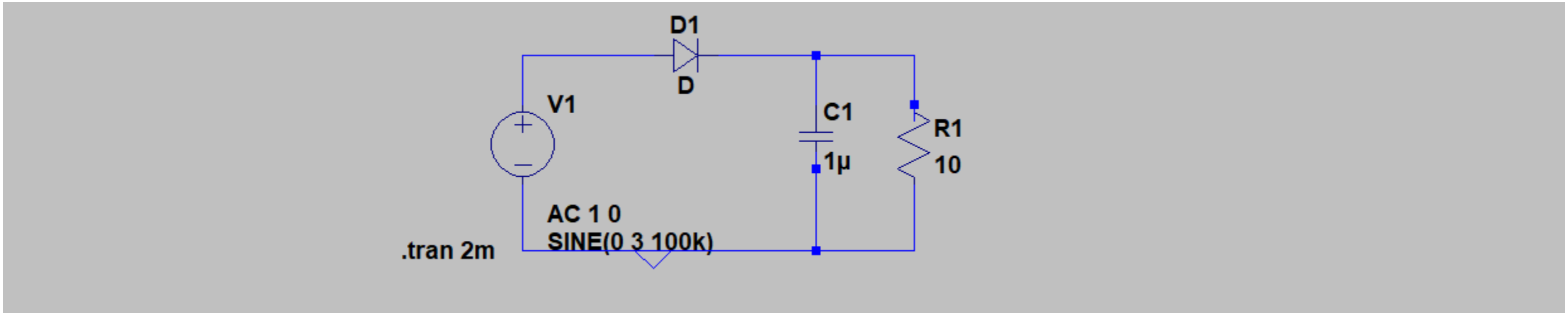




Diode-Cap rectifier with loading

- R=10





Diode-Cap rectifier with loading

- R=1K

